Deliverable #3

SE 3A04: Software Design II – Large System Design

Tutorial Number: T02
Group Number: G1
Group Members:

- Prutha Patel
- Kaiden Jessani
- Irfan Shahid
- Christopher Ludwig
- Mohammad Bilal

1 Introduction

1.1 Purpose

The purpose of this document is to provide further details about the architecture of the Smart City Environmental Monitoring & Alert System (SCEMAS), including state diagrams, sequence diagrams, and a detailed class diagram.

This document is intended for internal stakeholders at SCEMAS, including project managers, developers, domain experts, and investors. Familiarity with SCMEAS deliverables 1 and 2 is recommended, as they provide background information relevant to the architectural discussion.

1.2 System Description

An overview of the system description can be found in deliverable 2. This document is an extension of deliverable 2, providing further context of the system with state charts, sequence diagrams, and a detailed class diagram.

1.3 Overview

Section 2 consists of state charts for the controller classes: Sensor Service, Alerting Service, Account Service and Operational Module. Section 3 consists of relevant sequence diagrams of important events. Section 4 provides a detailed class diagram of the system.

2 State Charts for Controller Classes

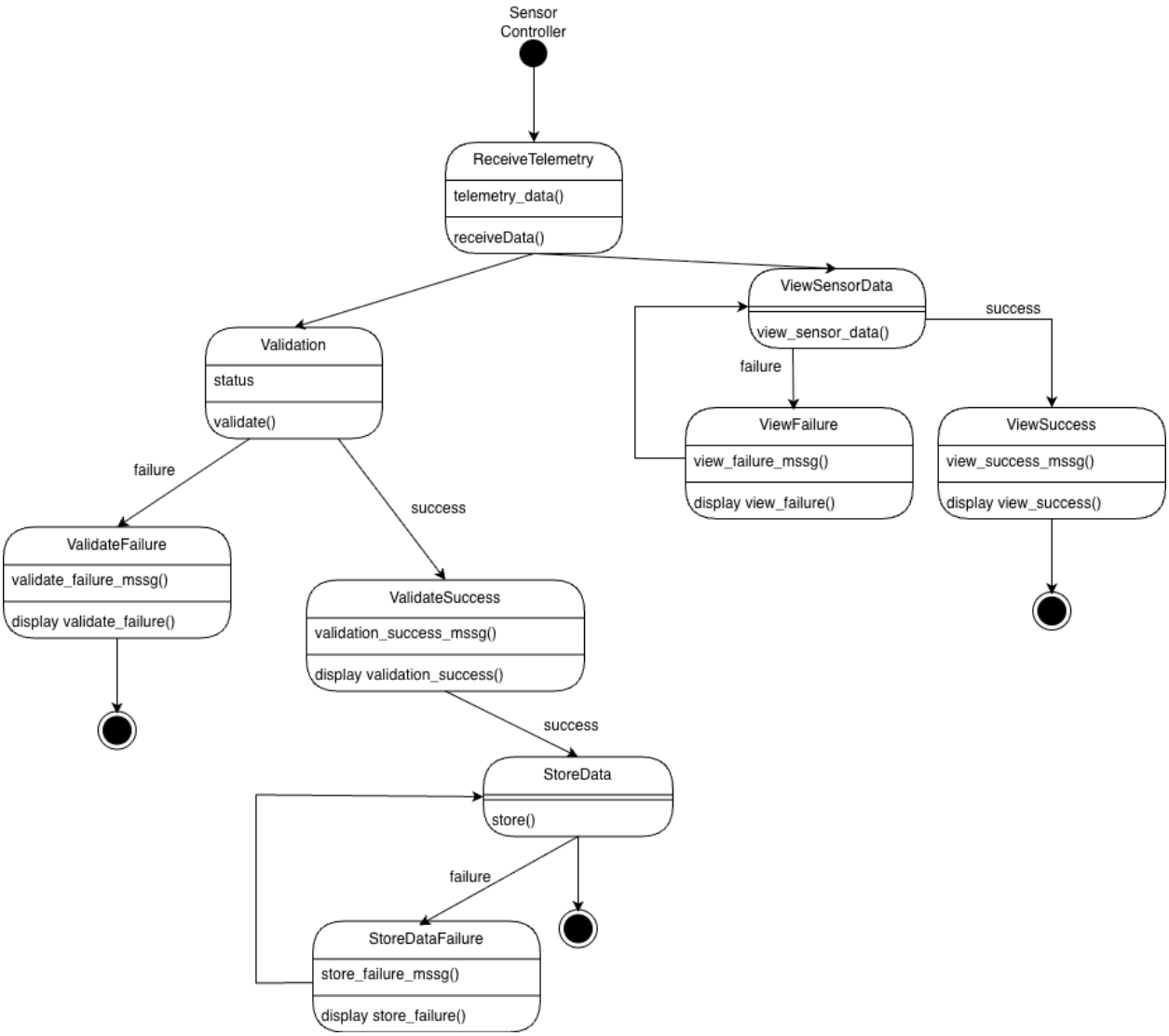

Figure 1: Sensor controller state diagram

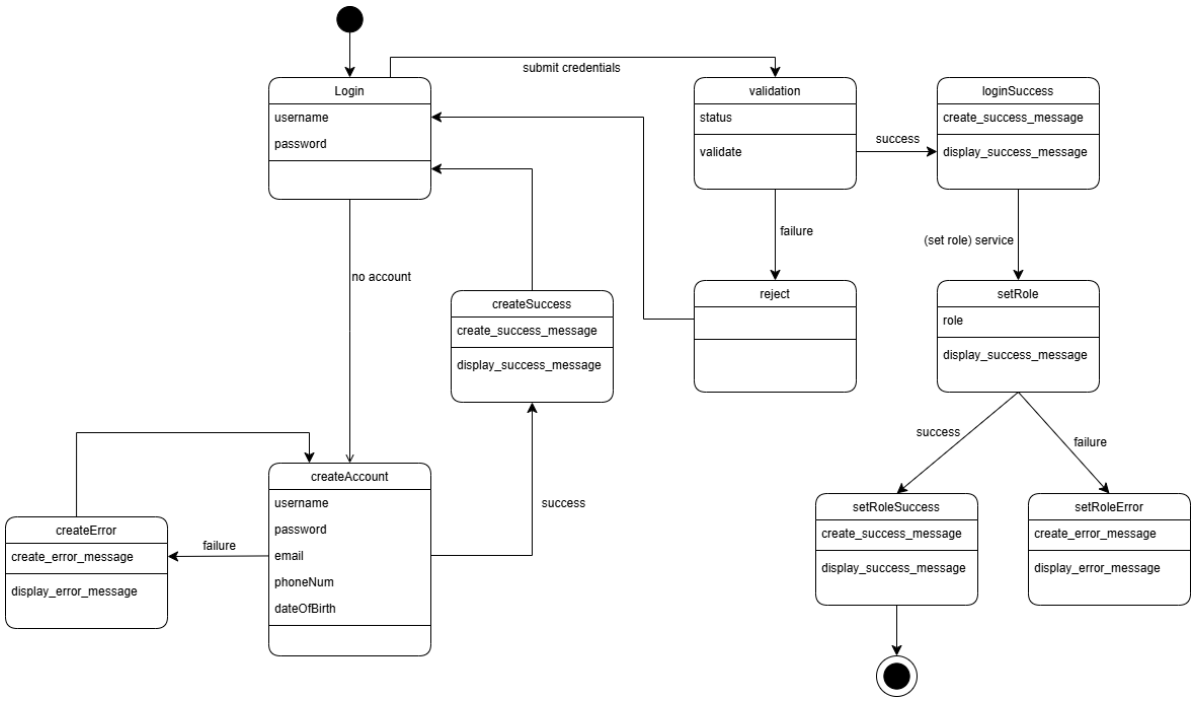

Figure 2: Account controller state diagram

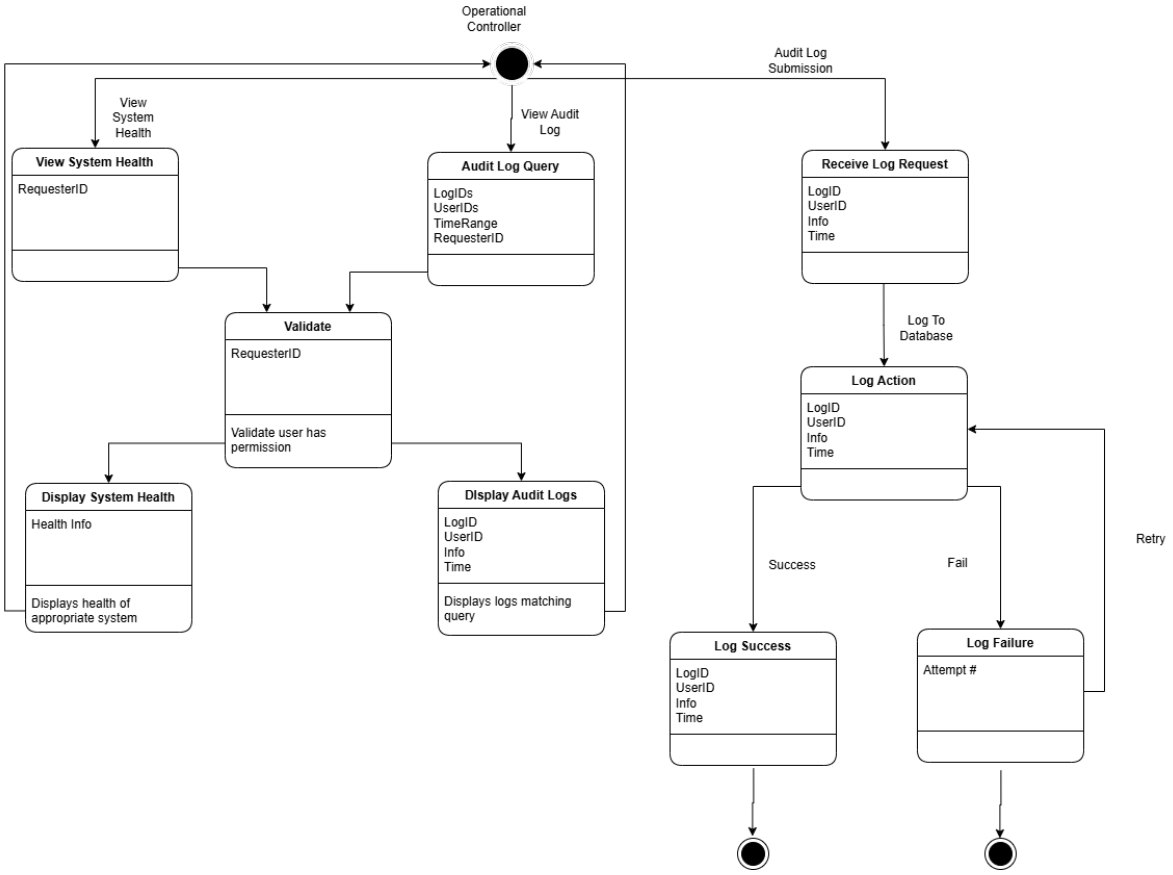

Figure 3: Operational controller state diagram

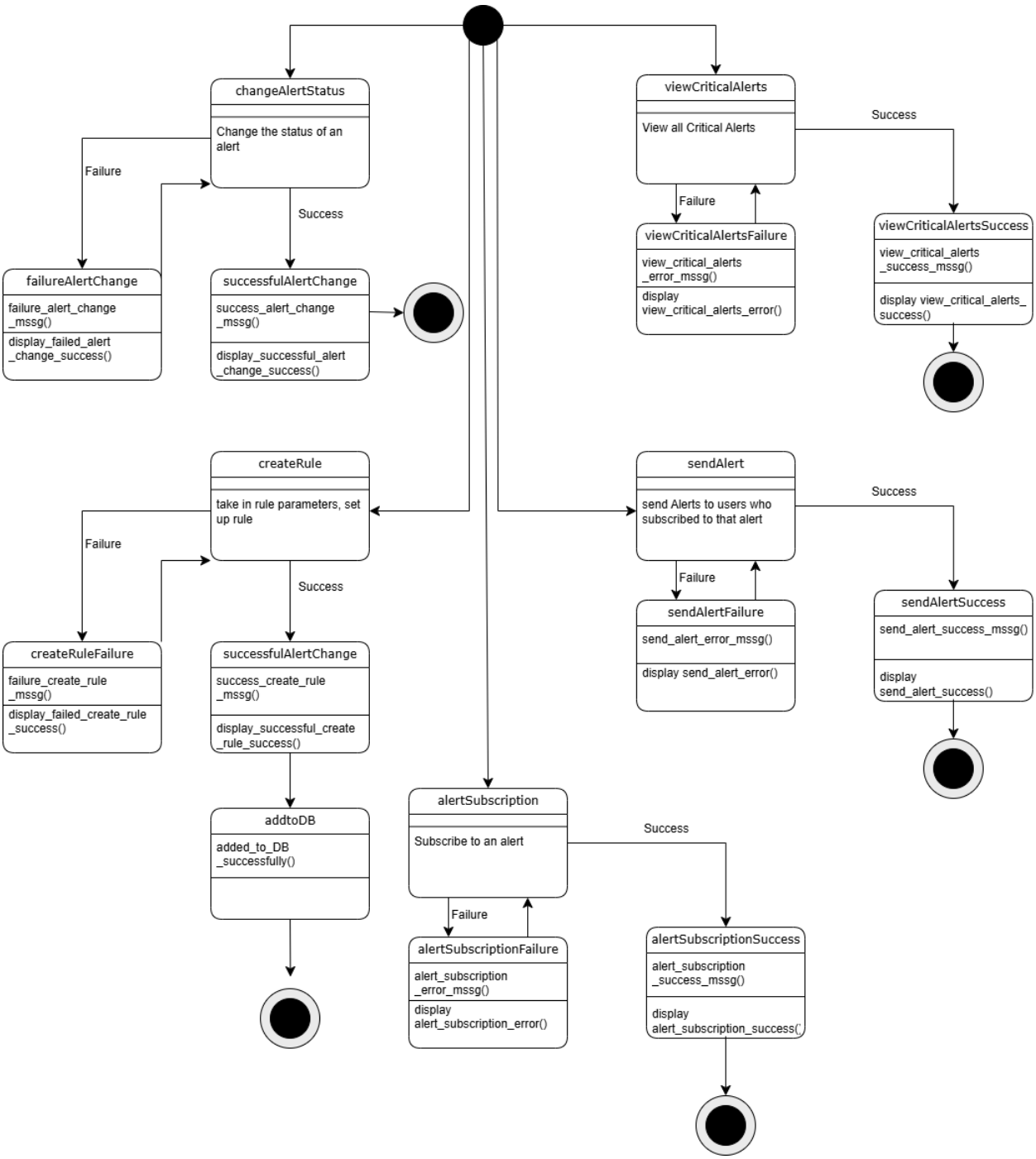

Figure 4: Alert controller state diagram

3 Sequence Diagrams

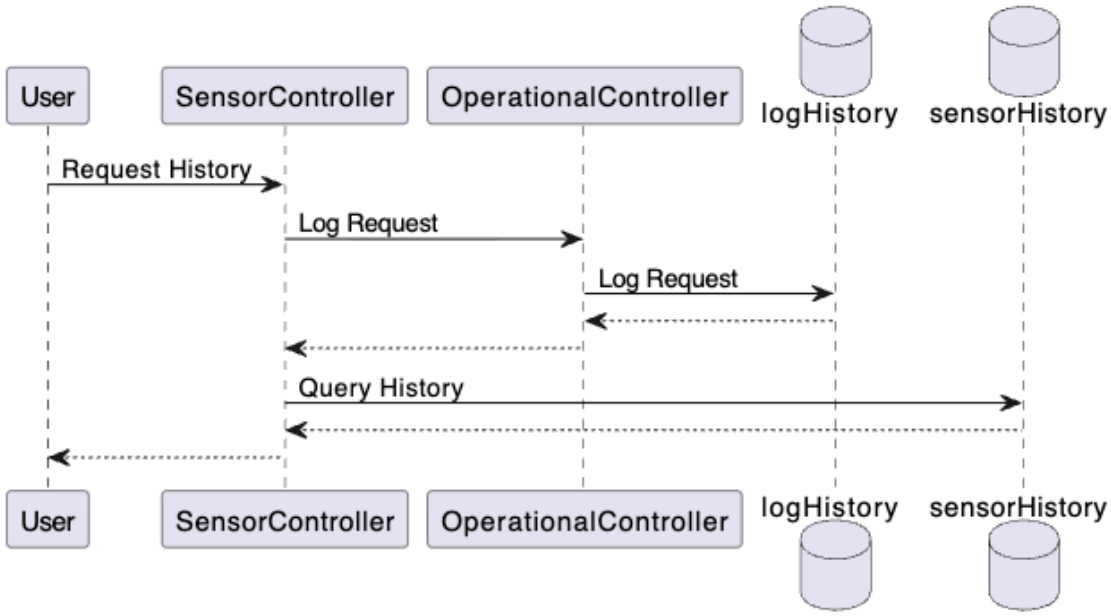

Figure 5: View historical sensor data sequence diagram

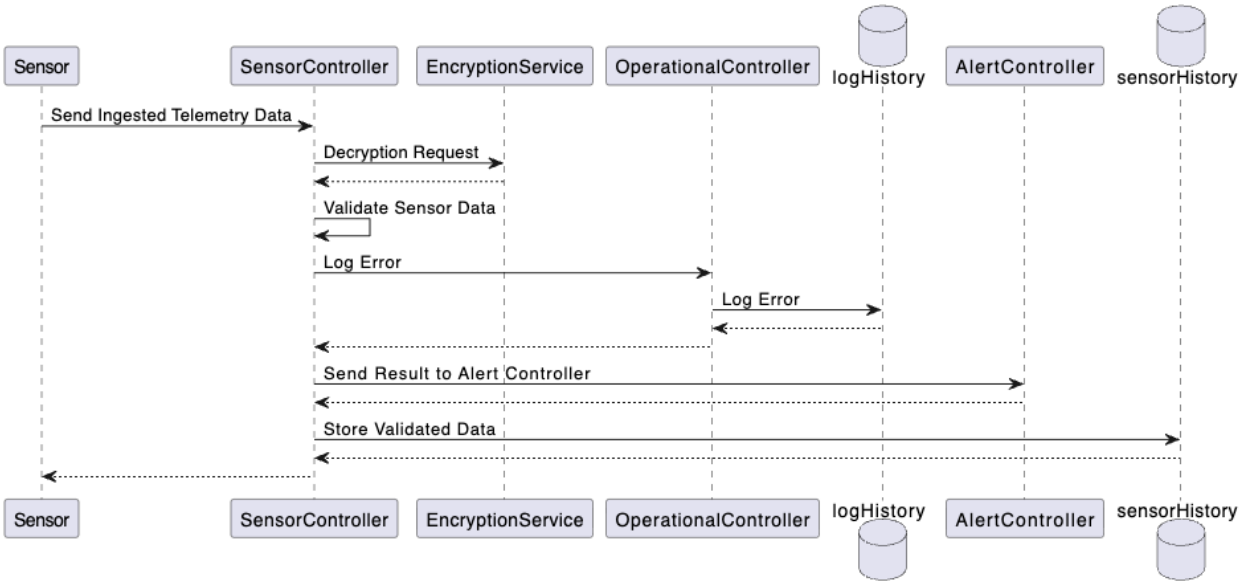

Figure 6: Ingest sensor data sequence diagram

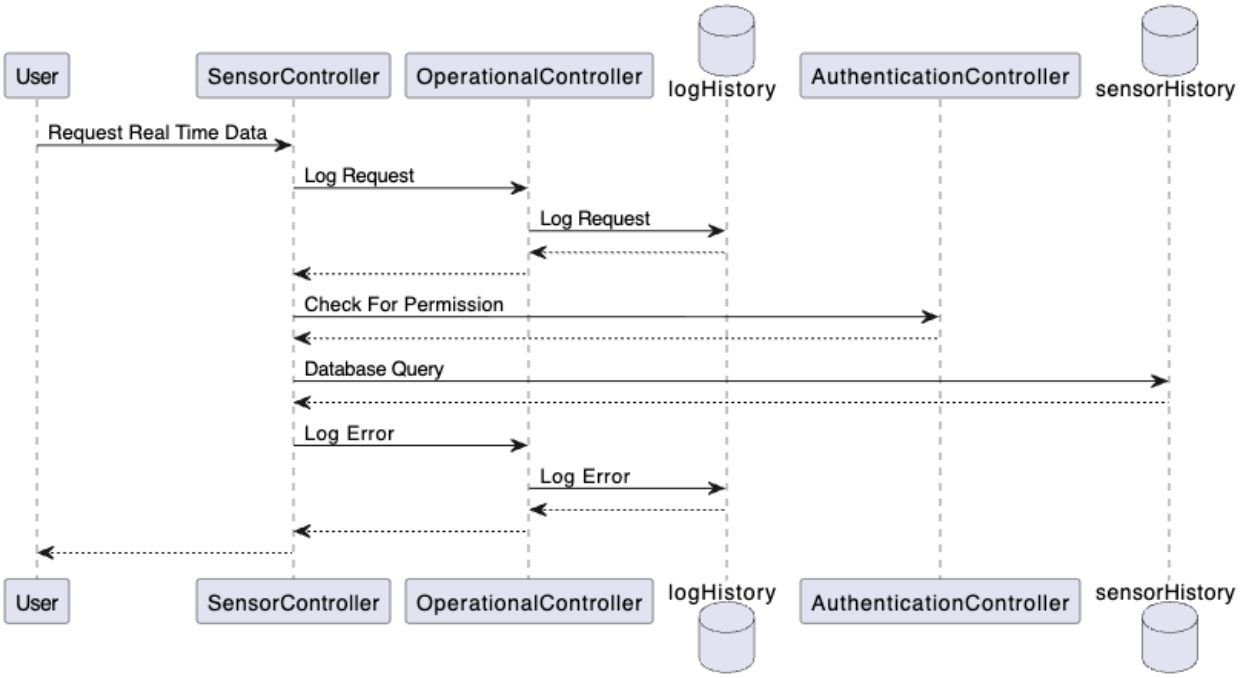

Figure 7: View real-time sensor data sequence diagram

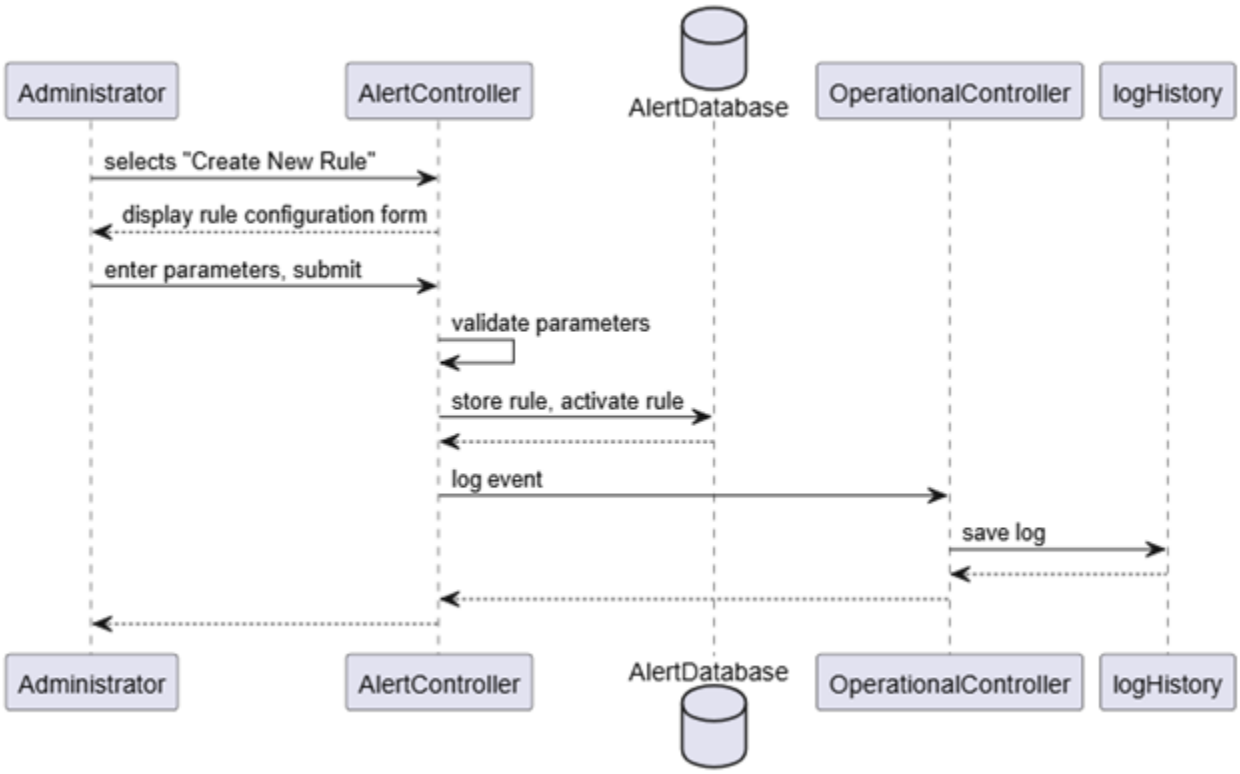

Figure 8: Define an alert rule sequence diagram

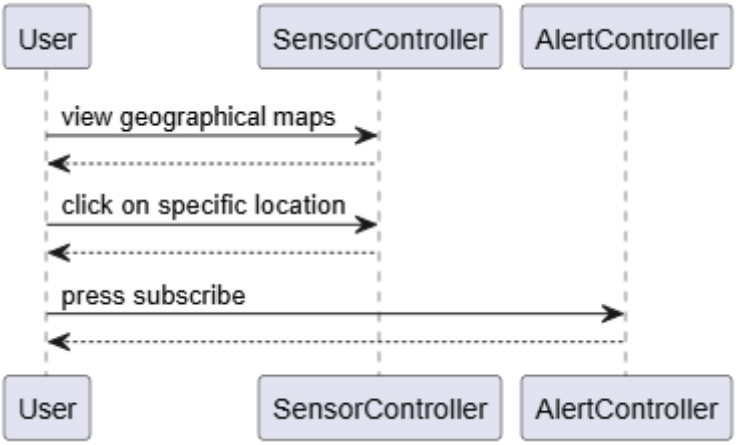

Figure 9: Subscribe to alerts sequence diagram

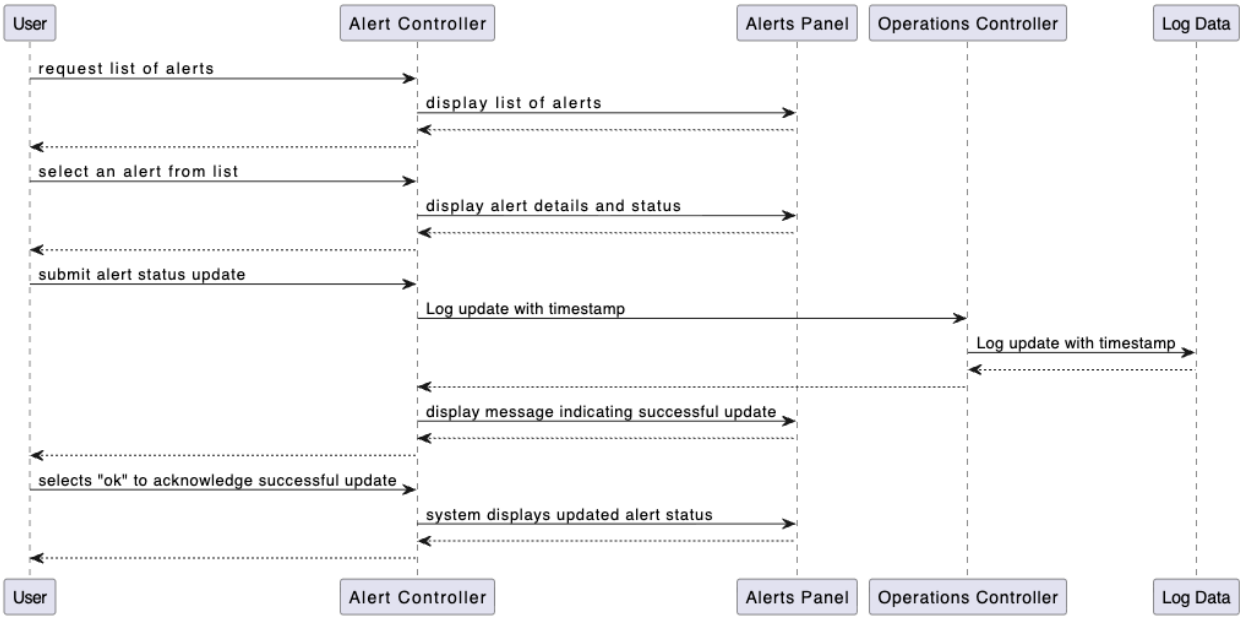

Figure 10: Manage alert status sequence diagram

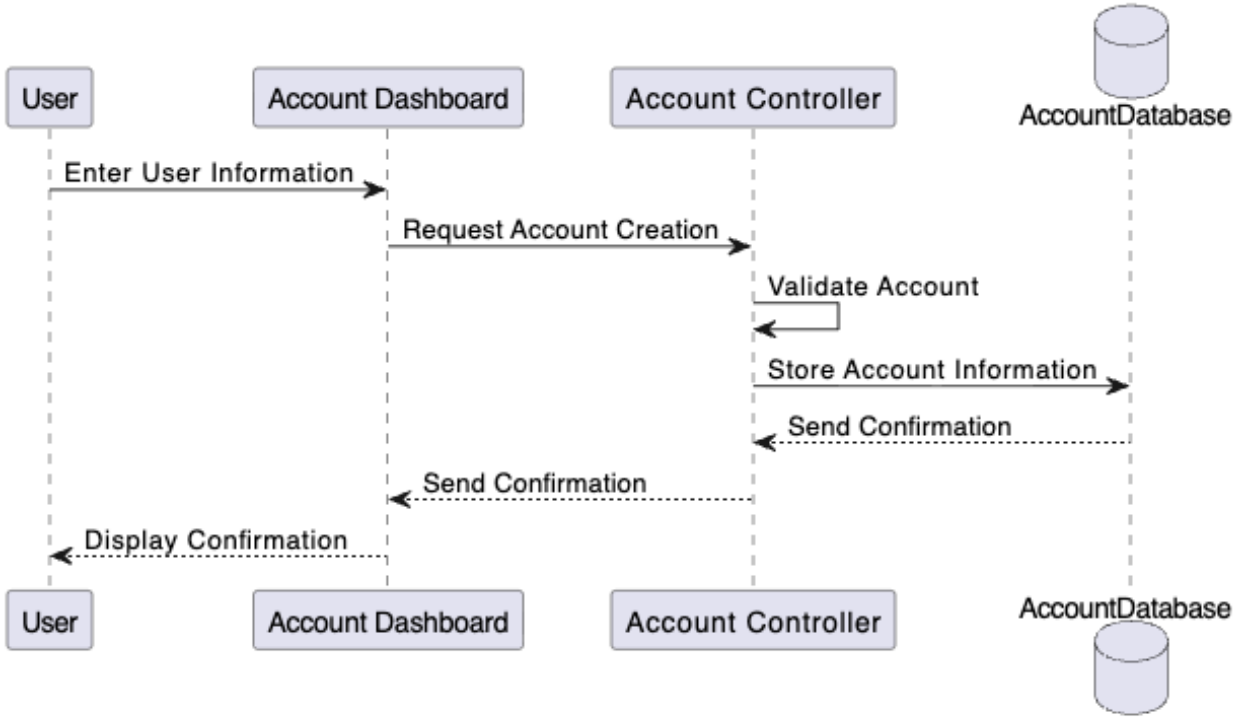

Figure 11: Create an account sequence diagram

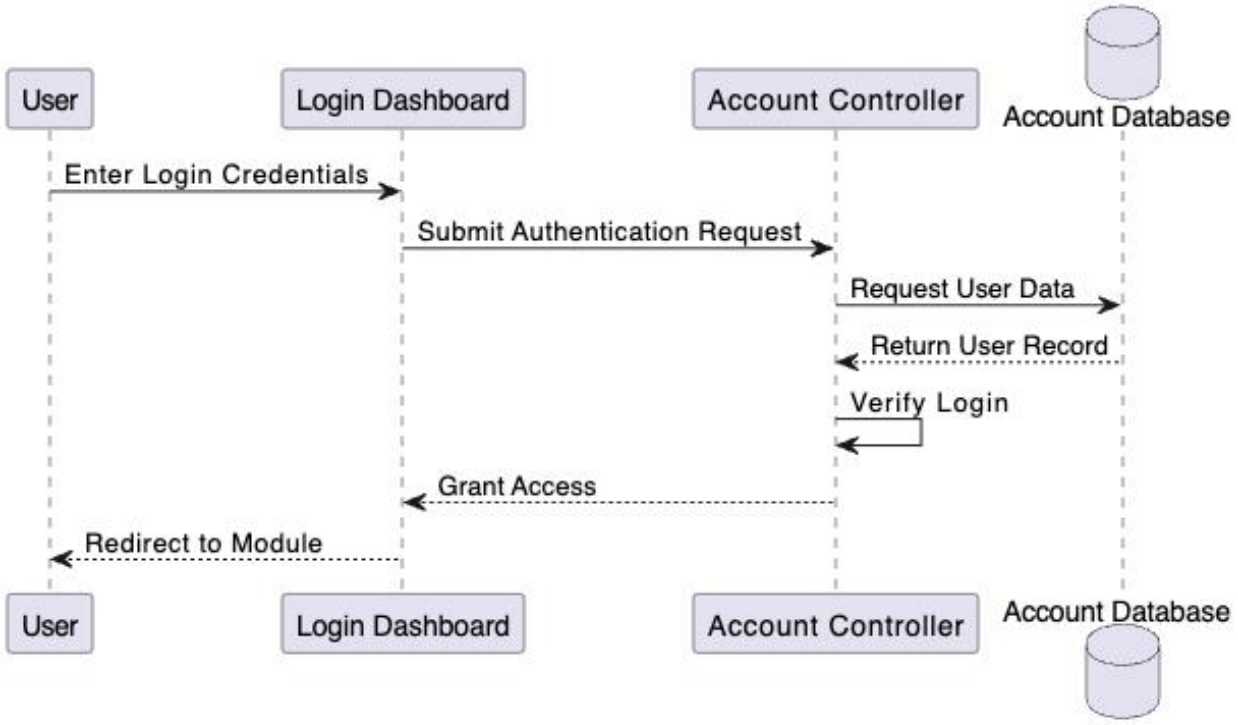

Figure 12: Login sequence diagram

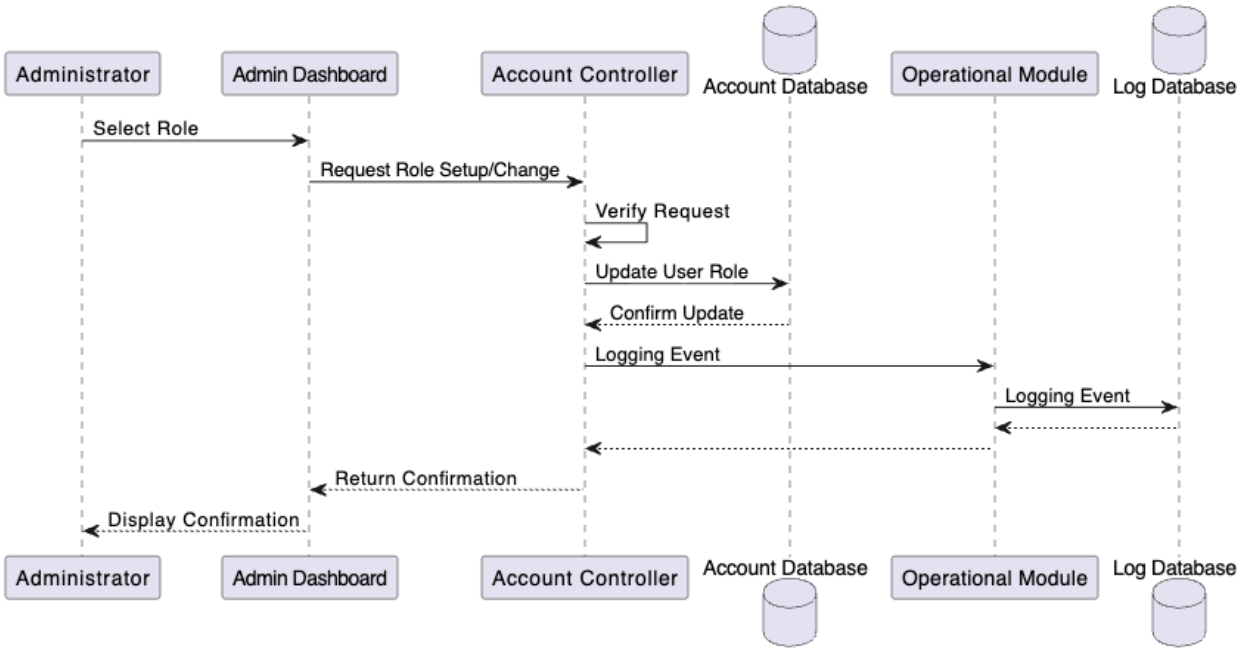

Figure 13: Change user role sequence diagram

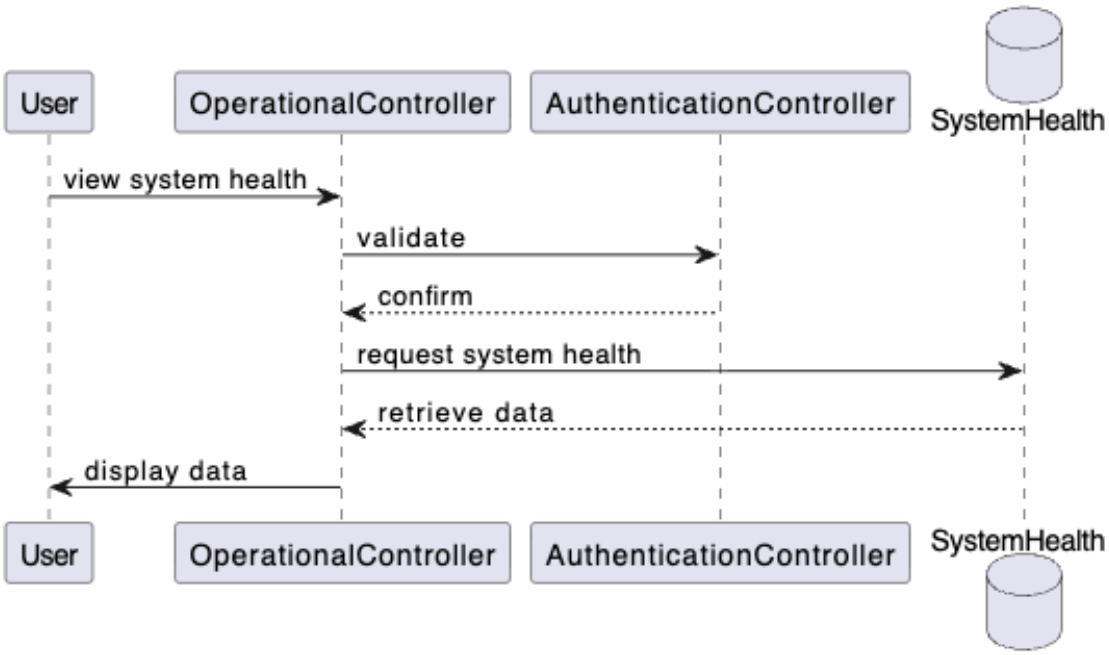

Figure 14: View system health sequence diagram

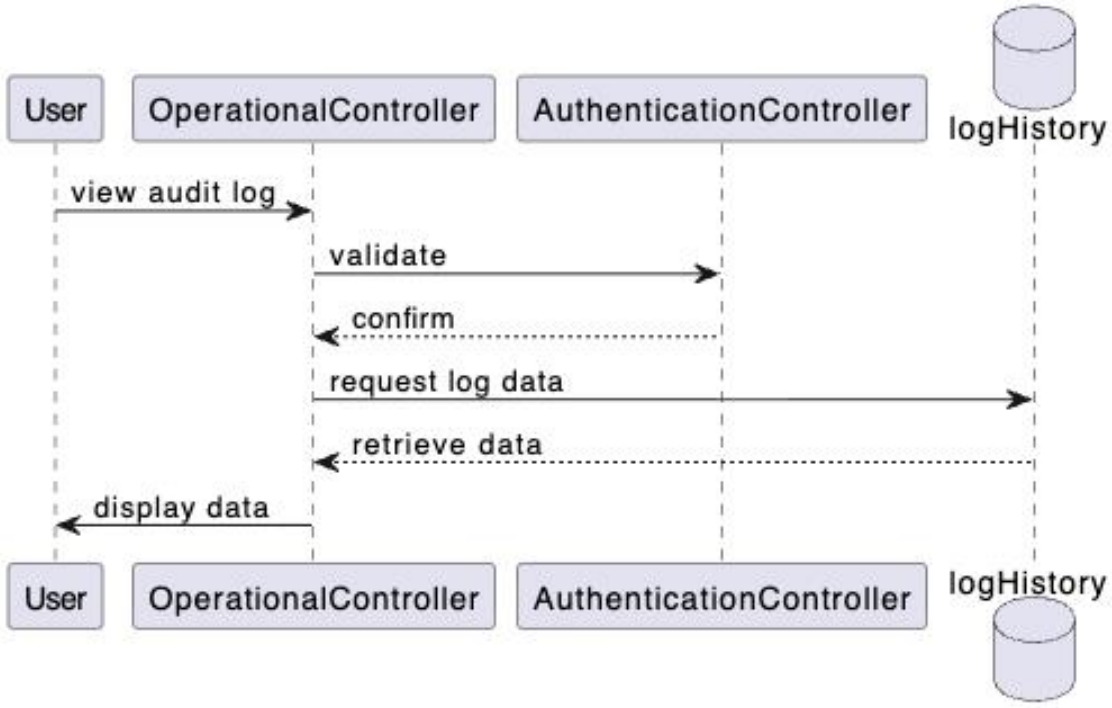

Figure 15: View audit logs sequence diagram

4 Detailed Class Diagram

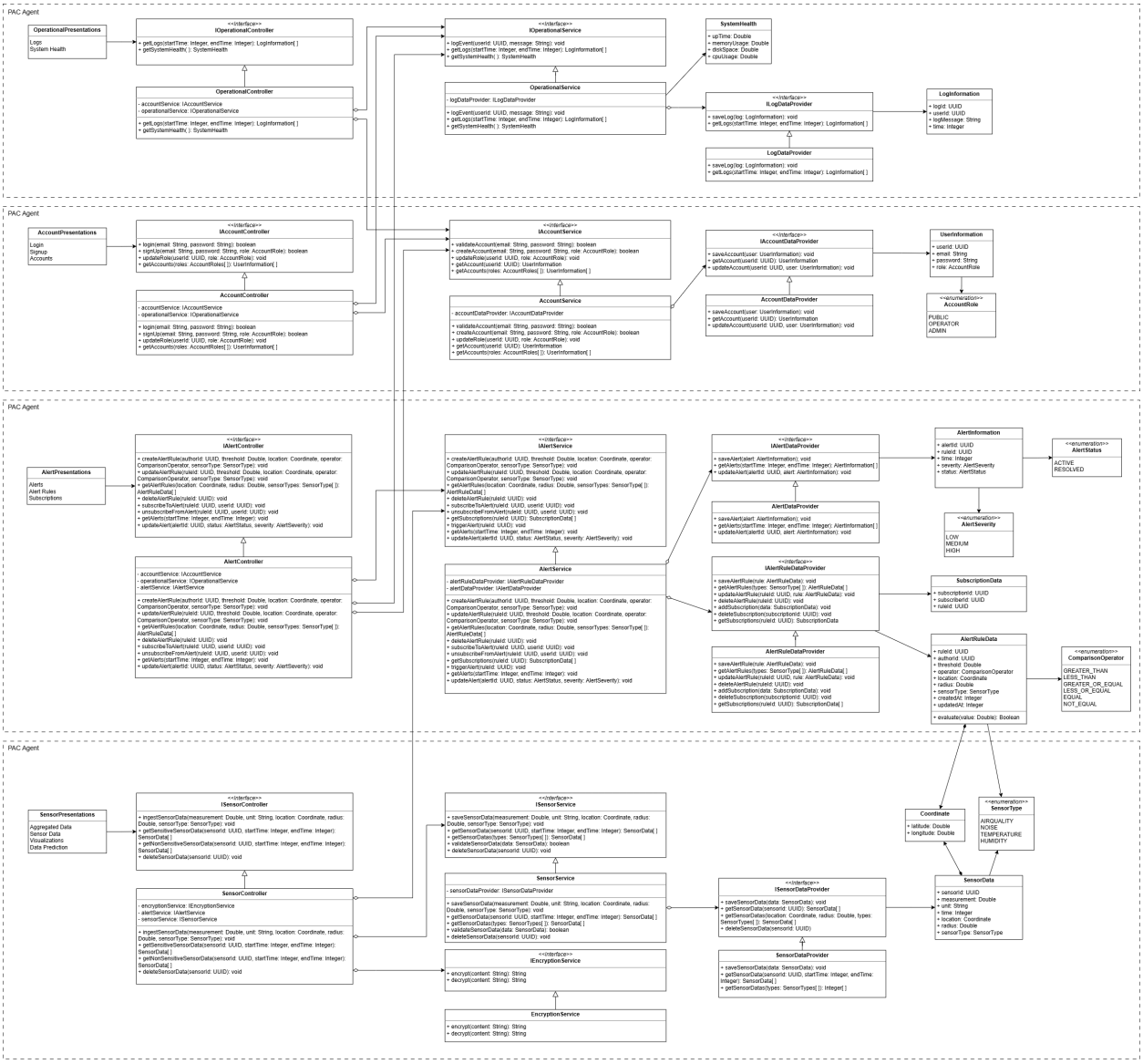

Figure 16: Overall class diagram

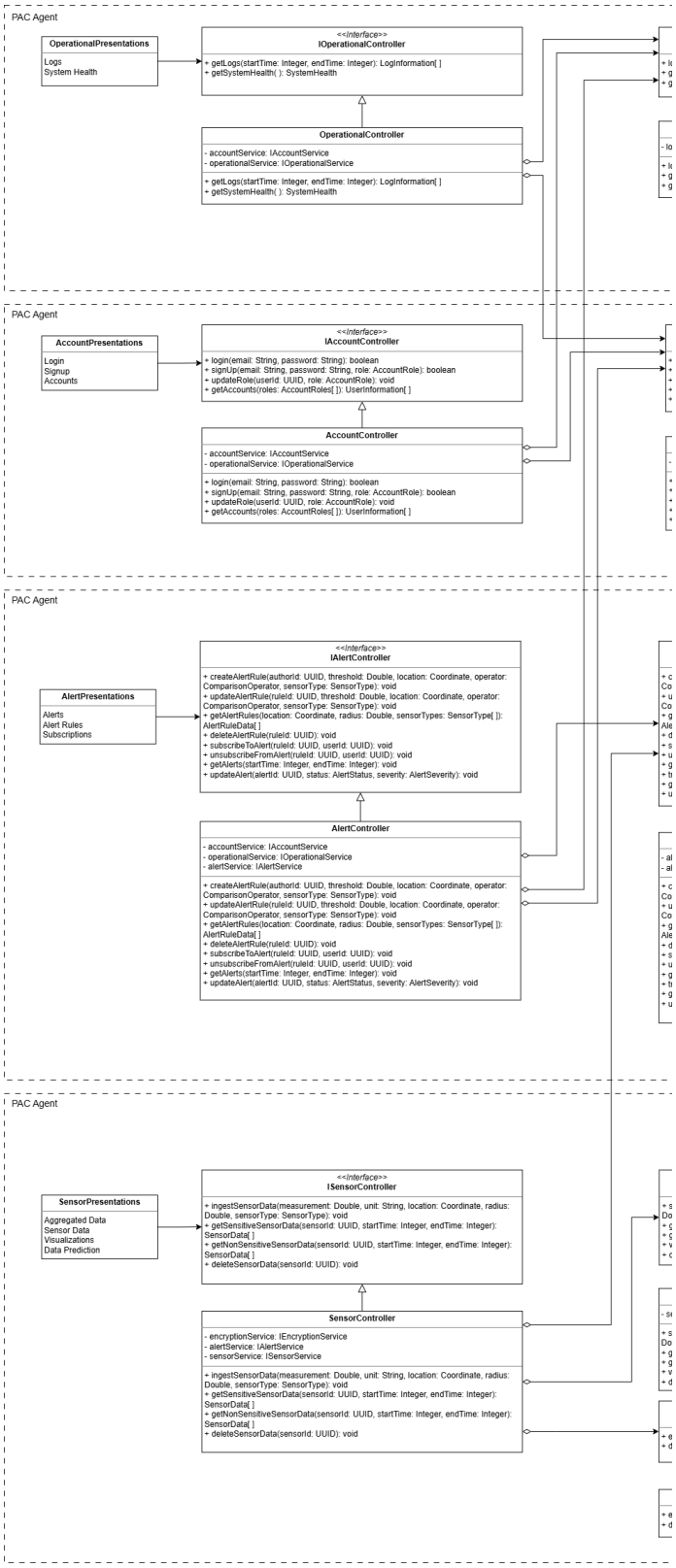

Figure 16.1: Left portion of the class diagram

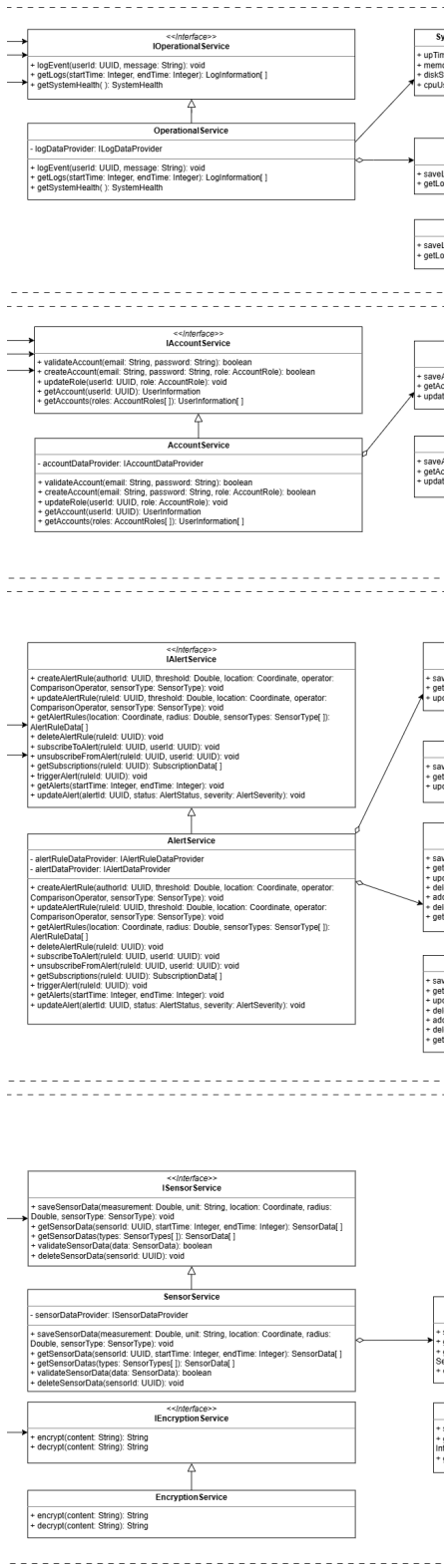

Figure 16.2: Middle portion of the class diagram

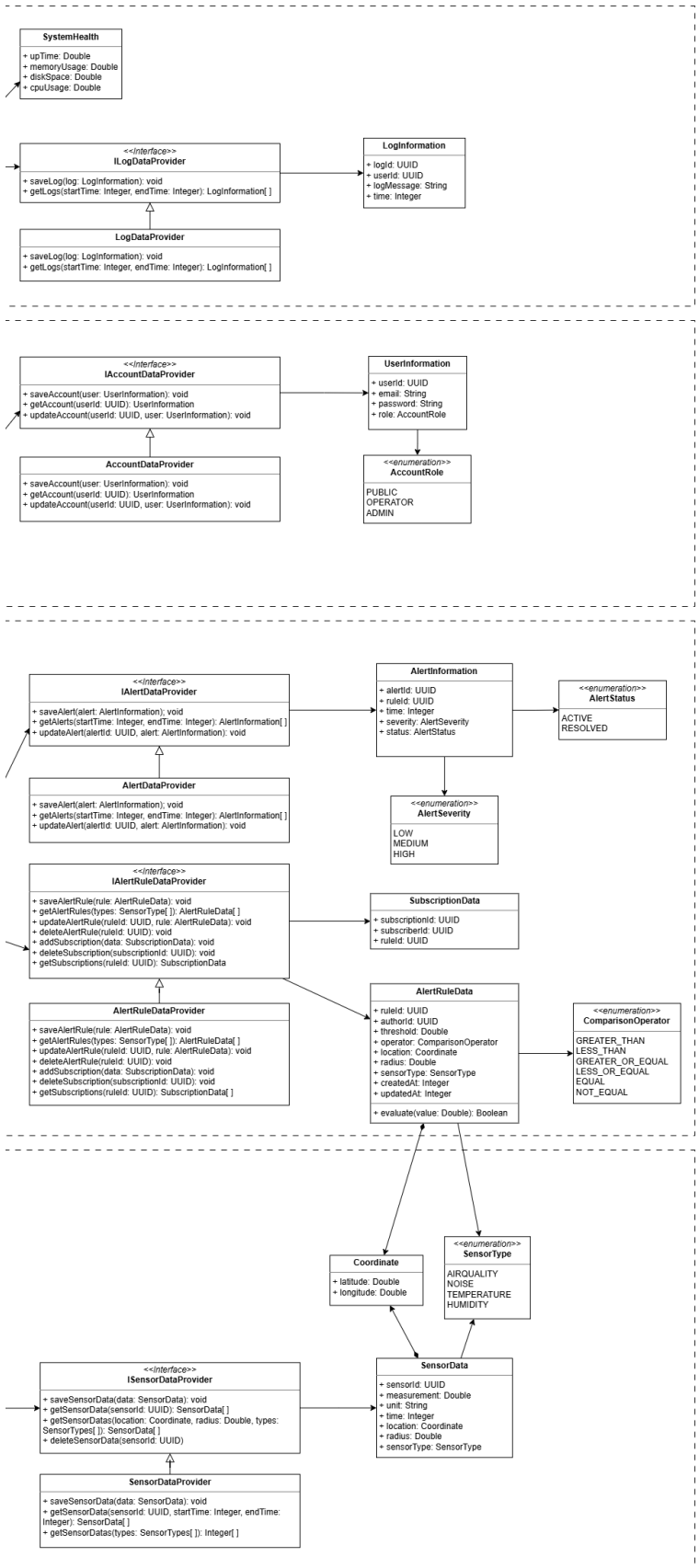

Figure 16.3: Right portion of the class diagram

A Division of Labour

Division of Labour sheet which indicates the contributions of each team member.

A.1 Bilal, Mohammad

- Section 4: Detailed Class Diagram
- Formatting and porting document to LaTeX

A.2 Jessani, Kaiden

- Section 2: Alert Controller State Chart
- Section 3: Define alert rule, subscribe to alerts, manage alerts

A.3 Ludwig, Christopher

- Section 1: Introduction
- Section 2: Operational Controller State Chart
- Section 3: View audit logs, view system health

A.4 Shahid, Irfan

- Section 2: Account Controller State Chart
- Section 3: Create account, login, change account role

A.5 Patel, Prutha

- Section 2: Sensor Controller State Chart
- Section 3: View historical data, ingest & store data, view real-time data